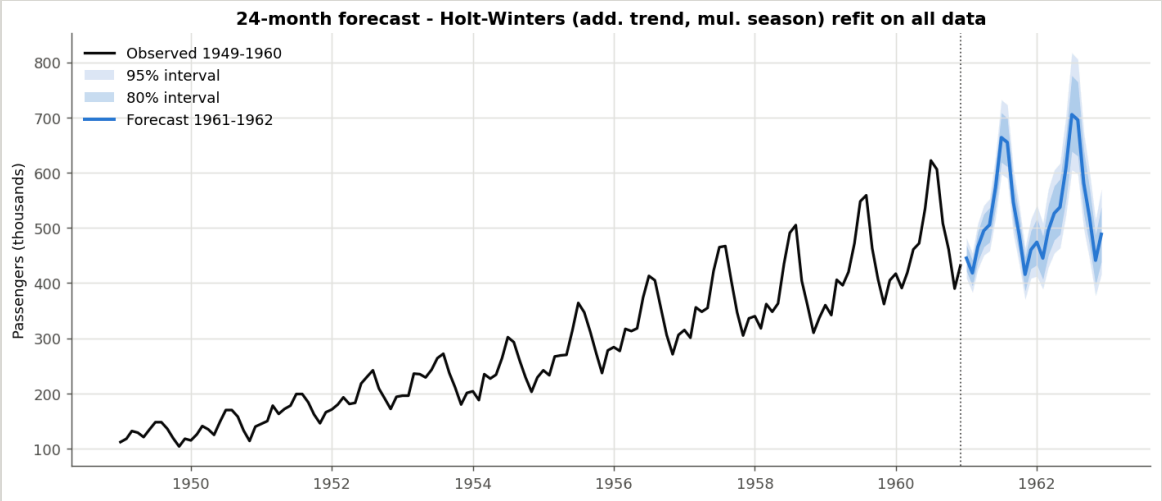
STATISTICAL METHODS (SM) - MINI PROJECT

Time Series Analysis and Forecasting of Monthly Airline Passengers (1949-1960)

Decomposition, stationarity testing, ARIMA / SARIMA / Holt-Winters and a 24-month forecast



Course	TY BTech - Statistical Methods
Submitted by	
Roll No.	
Tools	Python 3.12, pandas, statsmodels, matplotlib
Code	github.com/chhelu123/time-series-forecasting

1. Aim

To perform a complete **time series analysis** on a real monthly dataset: identify its components (trend, seasonality, irregular), test it for stationarity, identify a suitable model from its autocorrelation structure, compare classical forecasting methods on held-out data, check the best model's residuals, and forecast the next 24 months with confidence intervals.

2. Description

A *time series* is a sequence of observations recorded at equal time intervals. Unlike ordinary data, the observations are not independent: this month's value depends on previous months. Time series analysis uses this dependence to describe the past and forecast the future.

This project works through the full Box-Jenkins workflow on the classic *Air Passengers* series:

1. Load and clean the data, compute summary statistics and plot the series
2. Rolling statistics, seasonal decomposition (additive, multiplicative, STL) and seasonal plots
3. Stationarity tests (ADF and KPSS), log transform, regular and seasonal differencing
4. ACF/PACF plots and model order identification
5. Train/test split (last 24 months as test), then seven forecasting models
6. Residual diagnostics (residual plot, histogram, Q-Q plot, ACF, Ljung-Box test)
7. Model evaluation with MAE, RMSE and MAPE
8. Refit the best model on all data and forecast 1961-1962
9. Conclusions

3. Dataset

Item	Detail
Name	International Airline Passengers (Box & Jenkins "Series G")
Variable	Monthly total of international airline passengers, in **thousands**
Period	January 1949 to December 1960
Observations	144 (12 years x 12 months), no missing values
File	`data/AirPassengers.csv` (columns `Month`, `Passengers`)
Original source	Box, G.E.P., Jenkins, G.M. & Reinsel, G.C. *Time Series Analysis: Forecasting and Control*, Series G (1976). Also bundled with R as `datasets::AirPassengers`.
Download URL	https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv

It was chosen because it is the textbook example of a series with **both trend and seasonality**, and because the variance grows with the level, which makes it a good case for log transforms and multiplicative models.

Summary statistics (thousand passengers)

Statistic	Value	Statistic	Value
Mean	280.30	Minimum	104 (Nov 1949)
Standard deviation	119.97	Maximum	622 (Jul 1960)
Median	265.50	Skewness	0.58
Q1 / Q3	180.0 / 360.5	Coefficient of variation	42.8 %

The yearly total rose from 1,520 thousand (1949) to 5,714 thousand (1960), a 3.8x increase. Growth was positive every year (3.4 % to 21.8 % per year). The year-wise table is saved in `outputs/yearly_statistics.csv`.

4. Methodology (theory)

4.1 Components of a time series

Component	Meaning	In this data
Trend (T)	long-term upward/downward movement	steady growth of air travel
Seasonality (S)	pattern repeating with a fixed period (here 12 months)	summer holiday peak (Jul-Aug)
Cyclical (C)	rises and falls longer than a year with no fixed period	not visible in 12 years
Irregular / residual (R)	random noise left after removing the above	small

- **Additive model:** Y = T + S + R. Use it when the seasonal swing is constant.
- **Multiplicative model:** Y = T x S x R. Use it when the seasonal swing grows with the level. Taking logs turns a multiplicative model into an additive one: log Y = log T + log S + log R.
- **STL** (Seasonal-Trend decomposition using LOESS) is a robust, non-parametric decomposition that lets the seasonal pattern change slowly over time.

4.2 Stationarity

A series is (weakly) **stationary** if its mean, variance and autocovariance do not change over time. ARIMA models require a stationary series. Common fixes:

- **Log transform** to stabilise variance that grows with the level
- **First difference** (Y_t - Y_(t-1)) to remove a trend
- **Seasonal difference** (Y_t - Y_(t-12)) to remove a stable seasonal pattern

4.3 ADF and KPSS tests

Test	Null hypothesis H0	Stationary if
Augmented Dickey-Fuller (ADF)	series has a unit root (non-stationary)	p < 0.05 (reject H0)
KPSS	series is stationary	p >= 0.05 (do not reject H0)

The two tests have opposite null hypotheses, so we use both. We call a series stationary only when ADF rejects **and** KPSS does not reject.

4.4 ACF and PACF

- **ACF** (autocorrelation function): correlation between Y_t and Y_(t-k) for each lag k.

- **PACF** (partial ACF): correlation at lag k after removing the effect of lags 1..k-1.
- Identification rules: AR(p) means the PACF cuts off after lag p and the ACF tails off. MA(q) means the ACF cuts off after lag q and the PACF tails off. Spikes at lags 12, 24, ... point to seasonal AR (P) or MA (Q) terms.
- Bars outside the band +/- 1.96/sqrt(n) are significant at the 5 % level.

4.5 ARIMA and SARIMA

ARIMA(p, d, q): an AutoRegressive model of order p on the series differenced d times, plus Moving-Average terms of order q:

```
(1 - phi_1 B - ... - phi_p B^p) (1 - B)^d Y_t = (1 + theta_1 B + ... + theta_q B^q) e_t
```

where B is the backshift operator (B Y_t = Y_(t-1)) and e_t is white noise.

SARIMA(p, d, q)(P, D, Q)[s] adds seasonal AR/MA polynomials in B^s and seasonal differencing (1 - B^s)^D. Here s = 12. The well-known **"airline model"** is SARIMA(0,1,1)(0,1,1)[12]. Orders are chosen with the **AIC** (Akaike Information Criterion = 2k - 2 ln L), which balances goodness of fit against the number of parameters k. Lower is better.

4.6 Exponential smoothing and Holt-Winters

- **Simple exponential smoothing (SES):** level_t = alpha Y_t + (1 - alpha) level_(t-1). It gives a flat forecast and suits data with no trend and no seasonality.
- **Holt-Winters (triple exponential smoothing / ETS):** three smoothing equations, for level (alpha), trend (beta) and seasonality (gamma). The multiplicative-seasonal version forecasts (level + h x trend) x seasonal index.
- **Baselines:** *naive* (repeat the last value), *seasonal naive* (repeat the value from the same month last year) and *moving average* (mean of the last 12 months). A useful model must beat these.

4.7 Error metrics

Metric	Formula	Notes
MAE	(1/n) sum of abs(y_t - yhat_t)	average error, same unit as the data
RMSE	sqrt((1/n) sum of (y_t - yhat_t)^2)	penalises large errors more
MAPE	(100/n) sum of abs((y_t - yhat_t) / y_t)	percentage error, scale-free

4.8 Residual diagnostics

The residuals of an adequate model should be **white noise**: mean about 0, constant variance, roughly normal and uncorrelated. We check this with a residual plot, a histogram, a normal Q-Q plot, the residual ACF and the **Ljung-Box test** (H0: no autocorrelation up to lag h; p > 0.05 means the model is adequate).

5. How to run

```
git clone https://github.com/chhelu123/time-series-forecasting.git
cd time-series-forecasting
pip install -r requirements.txt

# Option 1: plain script, writes every figure and table to outputs/
python src/analysis.py

# Option 2: open the notebook (already executed, all outputs saved)
jupyter notebook time_series_analysis.ipynb
```

Run time is about 1 minute (most of it is the SARIMA grid search).

Project structure

```
time-series-forecasting/
|-- data/AirPassengers.csv            dataset
|-- src/analysis.py                   full pipeline as a script
|-- time_series_analysis.ipynb        same pipeline, executed notebook
|-- outputs/
|   |-- figures/*.png                 15 figures
|   |-- metrics.csv                   MAE / RMSE / MAPE of all models
|   |-- stationarity_tests.csv        ADF + KPSS results
|   |-- sarima_grid_search.csv        AIC of all 36 SARIMA models
|   |-- forecast_next_24_months.csv   forecast with 80 % / 95 % intervals
|   `-- ...                           summary stats, Ljung-Box tables
|-- report/report.html                printable report (source of REPORT.pdf)
|-- REPORT.pdf
`-- README.md                         this report
```

6. Key code snippets

Loading the data with a monthly DatetimeIndex

```
df = pd.read_csv(DATA)
df["Month"] = pd.to_datetime(df["Month"], format="%Y-%m")
df = df.drop_duplicates("Month").set_index("Month").sort_index()
y = df["Passengers"].asfreq("MS").astype(float)   # MS = month-start frequency
```

`asfreq("MS")` fixes the frequency. Any missing month would appear as NaN, which lets us verify that the series is complete (it is: 0 missing values).

Decomposition

```
dec_add = seasonal_decompose(y, model="additive", period=12)
dec_mul = seasonal_decompose(y, model="multiplicative", period=12)
stl = STL(np.log(y), period=12, robust=True).fit()
```

ADF + KPSS in one function

```
def stationarity(series, name):
    s = series.dropna()
    adf_stat, adf_p, *_ = adfuller(s, autolag="AIC")
    kpss_stat, kpss_p, *_ = kpss(s, regression="c", nlags="auto")
    adf_ok, kpss_ok = adf_p < 0.05, kpss_p >= 0.05   # both must agree
    ...

d1_12 = np.log(y).diff().diff(12)    # log + first + seasonal difference
```

SARIMA order selection by AIC grid search (on log scale)

```
for p, q, P, Q in itertools.product(range(3), range(3), range(2), range(2)):
    res = SARIMAX(np.log(train), order=(p, 1, q),
                  seasonal_order=(P, 1, Q, 12)).fit(disp=False)
    sarima_grid.append({"order": (p, 1, q), "seasonal_order": (P, 1, Q, 12),
                        "AIC": res.aic})
# forecasts are transformed back with exp()
forecast = np.exp(sarima.get_forecast(24).predicted_mean)
```

Holt-Winters (additive trend, multiplicative seasonality)

```
hw = ExponentialSmoothing(train, trend="add", seasonal="mul",
                          seasonal_periods=12,
                          initialization_method="estimated").fit()
forecast = hw.forecast(24)
```

Evaluation metrics

```
def metrics(actual, pred):
    e = actual - pred
    return {"MAE": np.mean(np.abs(e)),
            "RMSE": np.sqrt(np.mean(e ** 2)),
            "MAPE (%)": 100 * np.mean(np.abs(e / actual))}
```

Residual diagnostics: Ljung-Box

```
lb = acorr_ljungbox(resid, lags=[6, 12, 18, 24], return_df=True)
```

7. Output and interpretation

7.1 The series

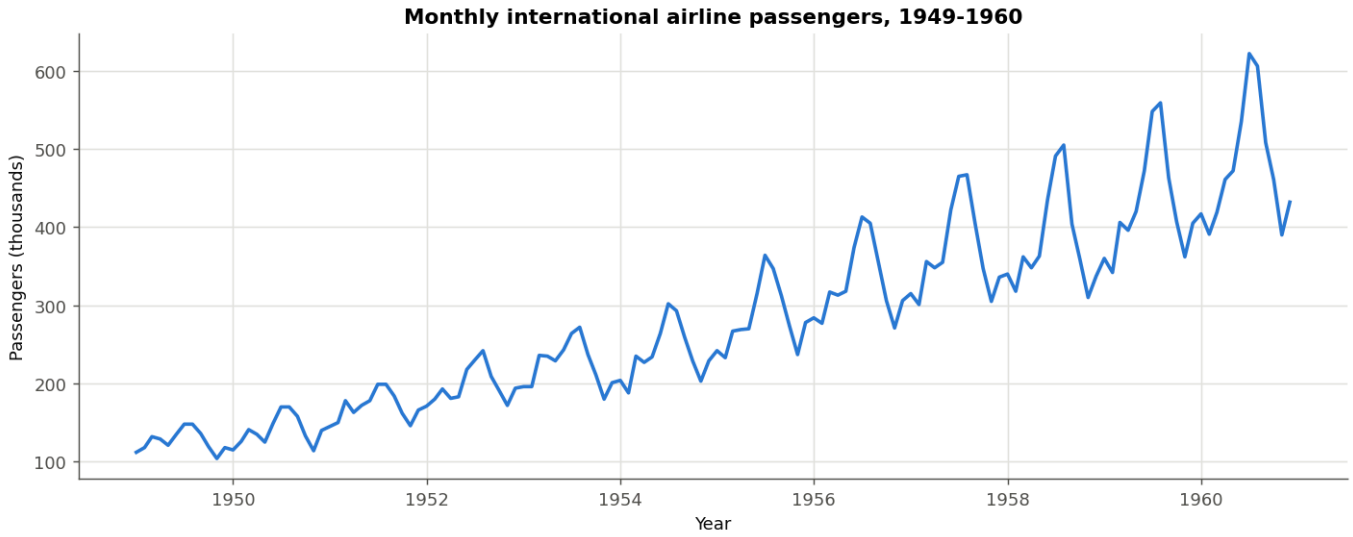


Figure 1: Time series

The series shows a clear **upward trend** and a **12-month seasonal cycle**, and the seasonal swings become wider as the level rises.

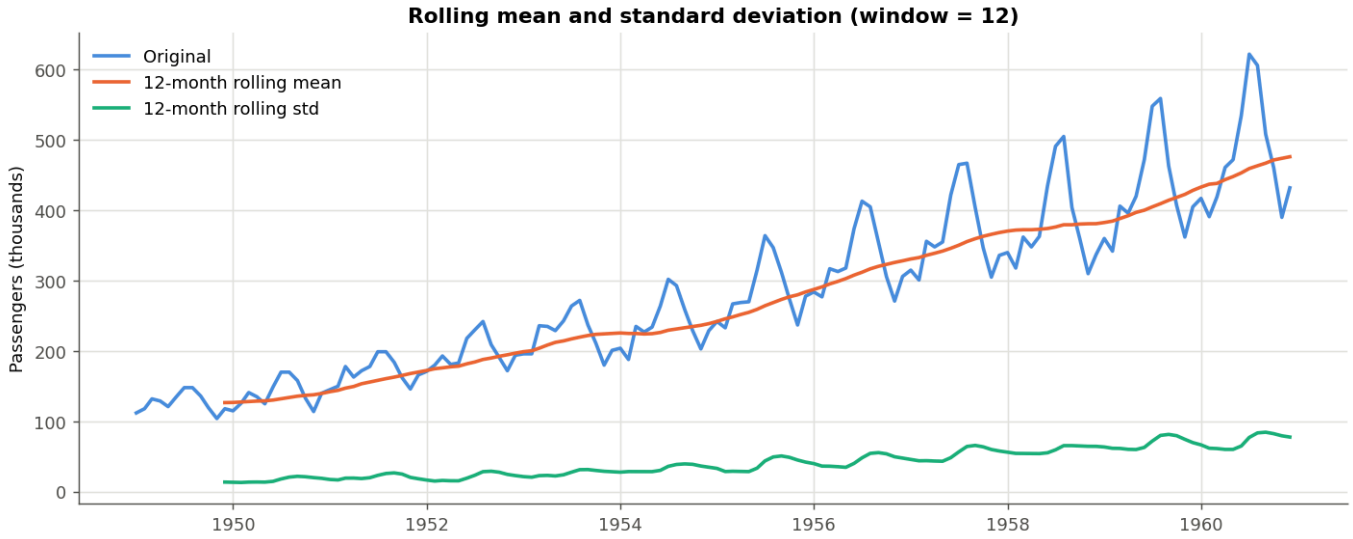


Figure 2: Rolling statistics

The 12-month rolling mean rises steadily. The rolling standard deviation rises from 13.7 (1949) to 77.7 (1960). Because both the mean and the variance change, the series is **not stationary**.

7.2 Decomposition

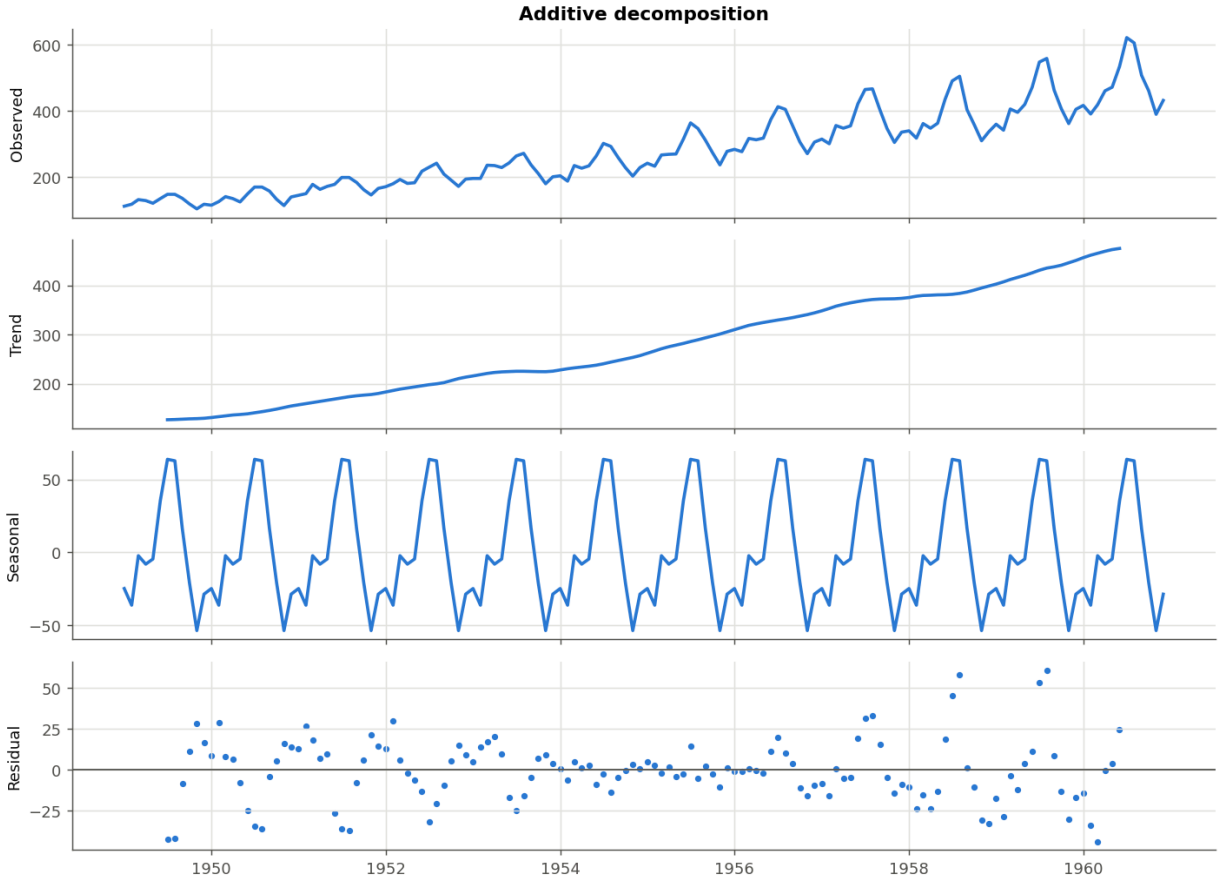


Figure 3: Additive decomposition

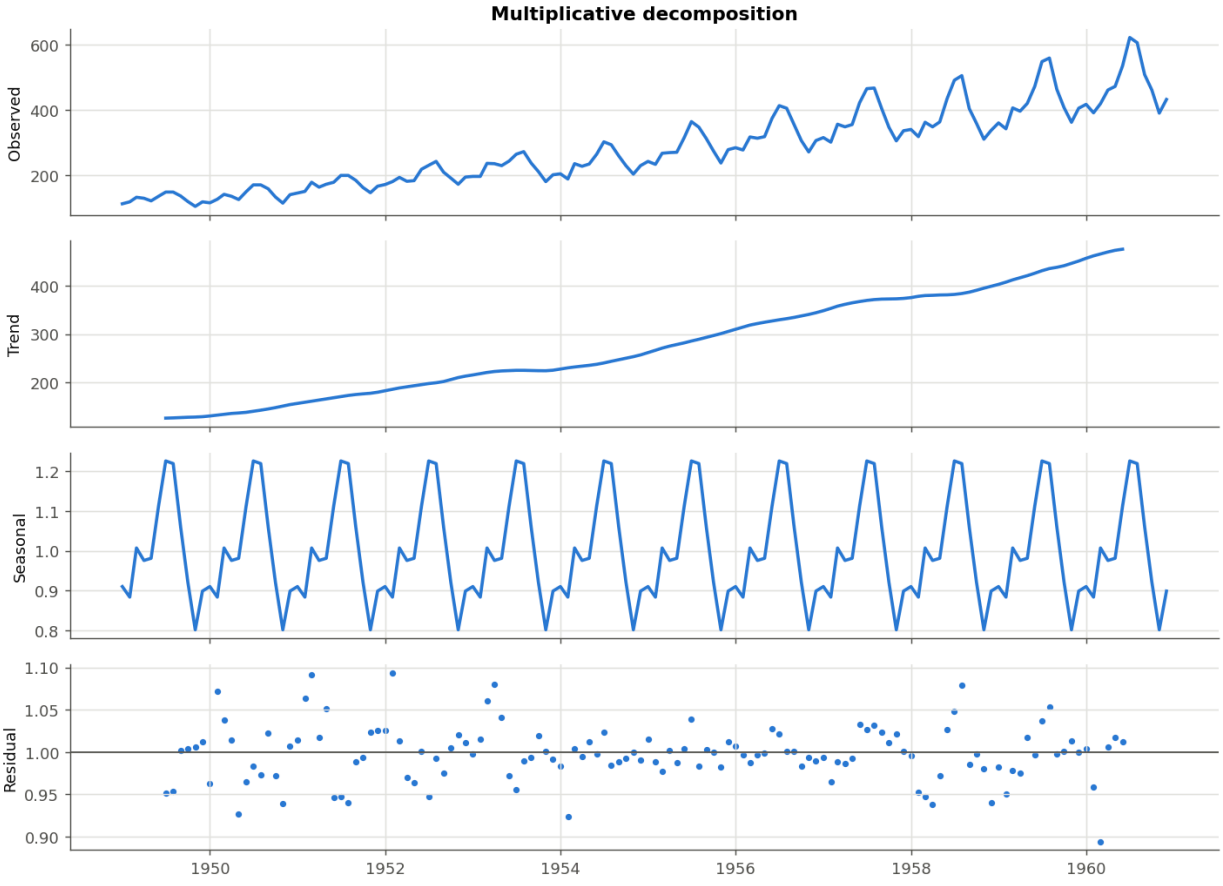


Figure 4: Multiplicative decomposition

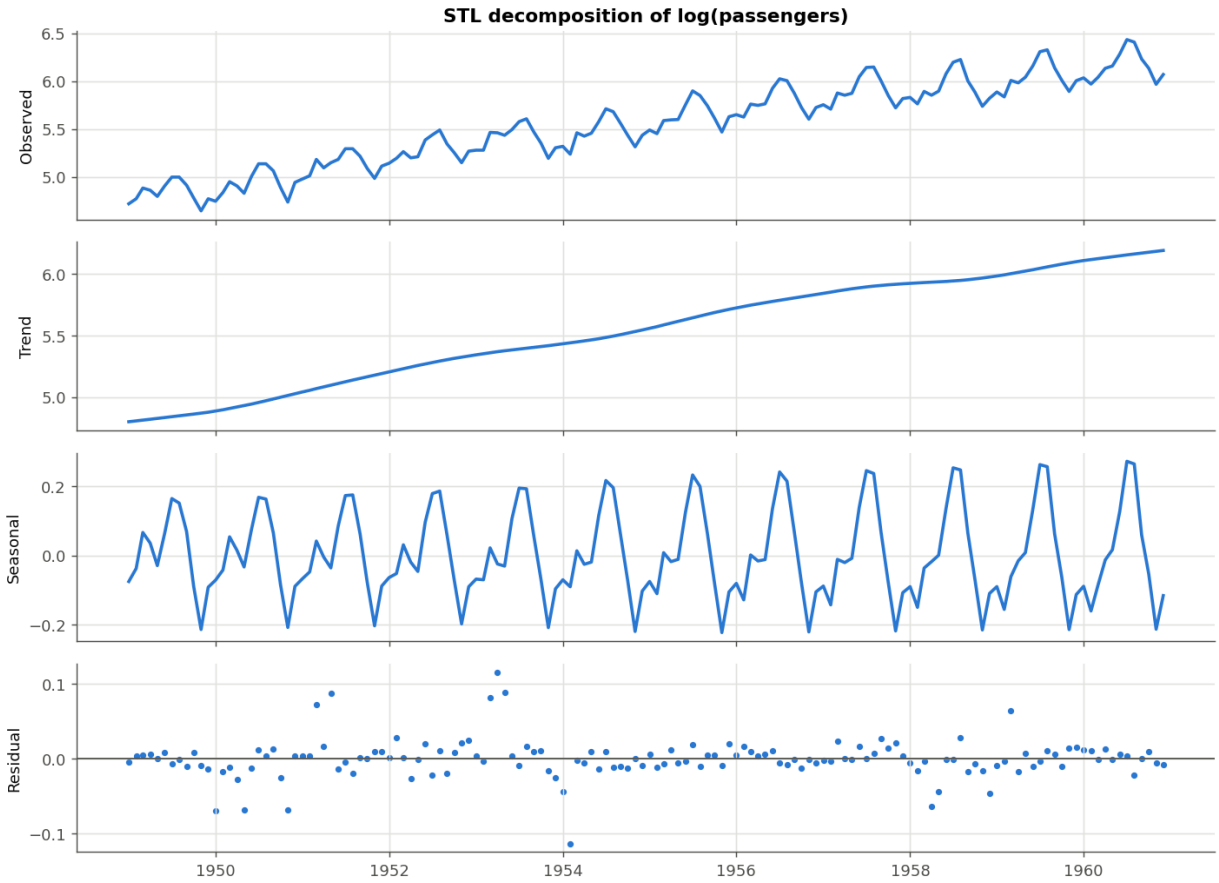


Figure 5: STL decomposition

- The **additive** residuals still contain a seasonal pattern that is large at both ends of the series (lag-1 autocorrelation 0.67). A fixed seasonal amount cannot describe swings that keep growing.
- The **multiplicative** residuals are smaller and less structured (lag-1 autocorrelation 0.40). The seasonal indices are Jul 1.227, Aug 1.220 and Nov 0.801: July traffic is about 23 % above an average month and November about 20 % below.
- **STL on log(Y)** gives a strength of trend of 0.996 and a strength of seasonality of 0.959 (both on a 0-1 scale), so both components are very strong.

7.3 Seasonality

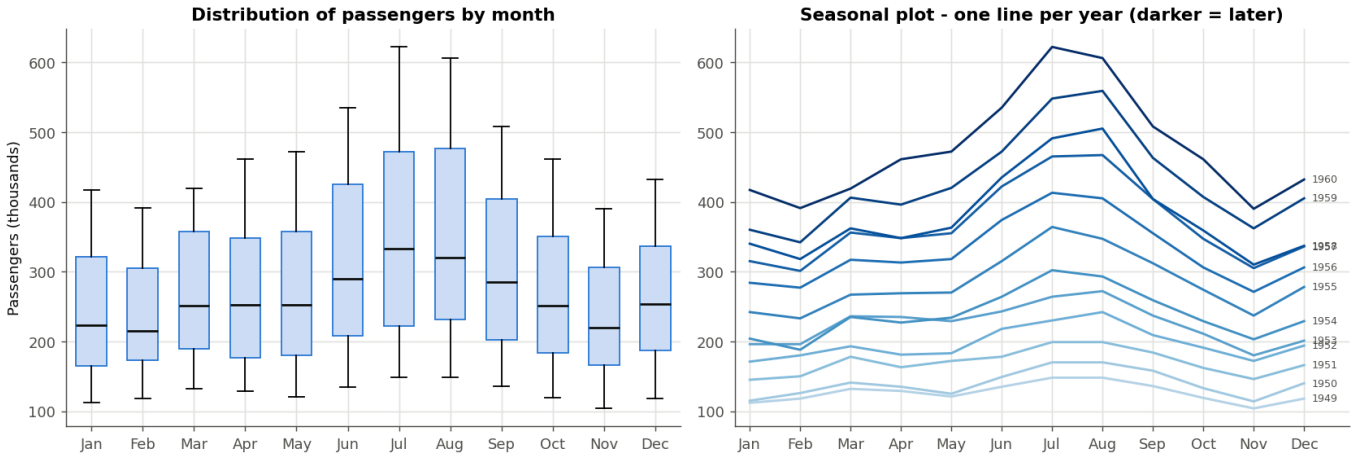


Figure 6: Seasonal box plot and seasonal plot

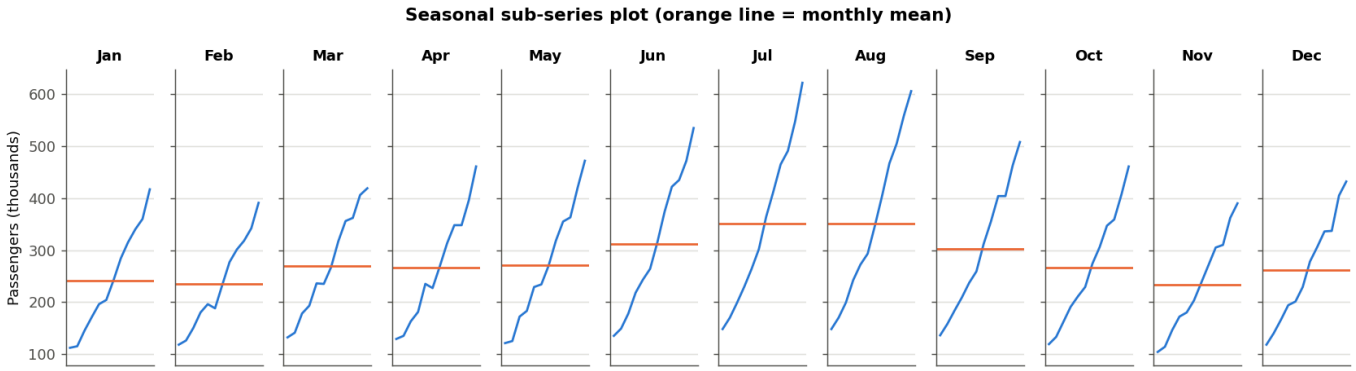


Figure 7: Seasonal sub-series

July and August are the busiest months in every year and November is the quietest. The year lines are stacked almost in order, and only 1957 and 1958 overlap (1958 had the weakest growth). The *shape* of the season is stable while its *level* rises. Each month's sub-series rises steadily (trend).

7.4 Stationarity tests

Series	ADF stat	ADF p	KPSS stat	KPSS p	Conclusion
Original Y	0.815	0.9919	1.651	0.01	Non-stationary
log(Y)	-1.717	0.4224	1.669	0.01	Non-stationary
log(Y), 1st difference	-2.717	0.0711	0.038	0.10	Inconclusive
log(Y), seasonal difference	-2.710	0.0724	0.292	0.10	Inconclusive
log(Y), 1st + seasonal difference	**-4.443**	**0.0002**	**0.073**	**0.10**	**Stationary**

(KPSS p-values are interpolated from a table and capped at 0.01 and 0.10.)

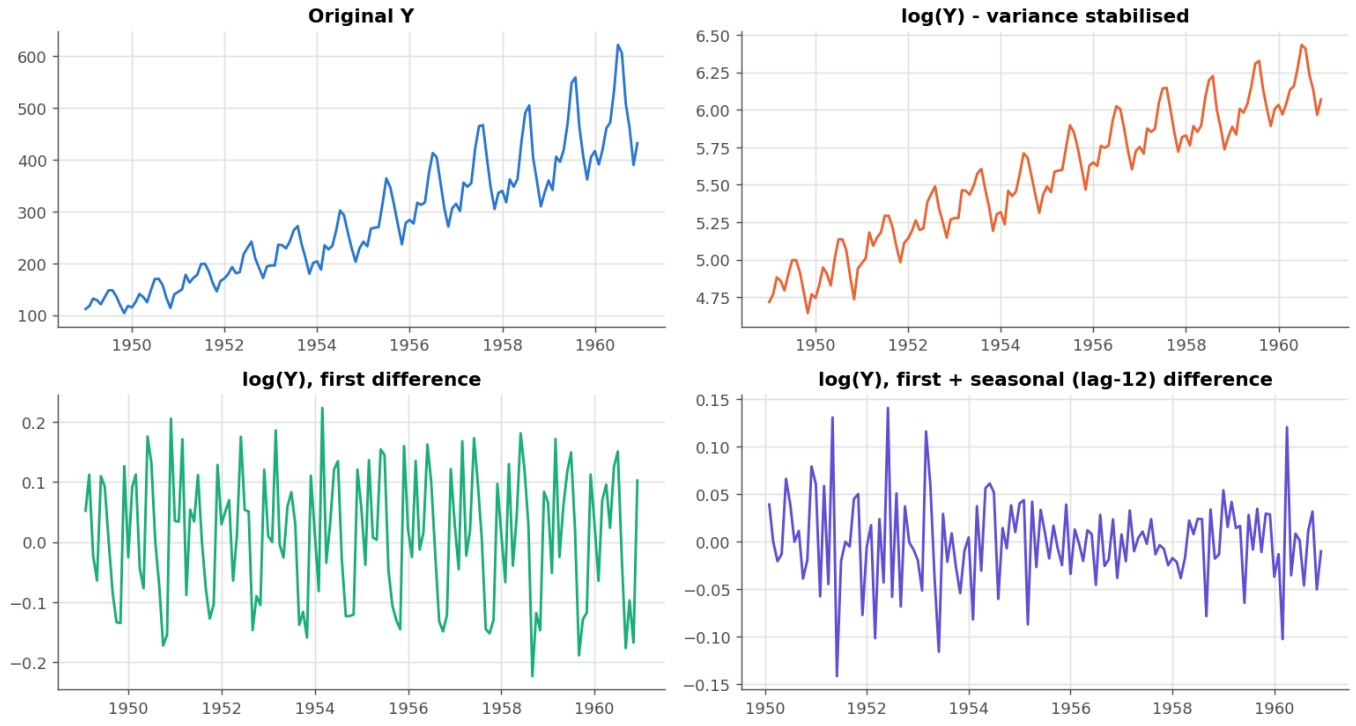


Figure 8: Transformations

The log transform makes the seasonal swings equal in size, but the trend remains. A single difference of either kind is not enough: ADF p is about 0.07. Only after one regular **and** one seasonal difference do both tests agree that the series is stationary. This gives **d = 1, D = 1, s = 12**.

7.5 ACF / PACF and model identification

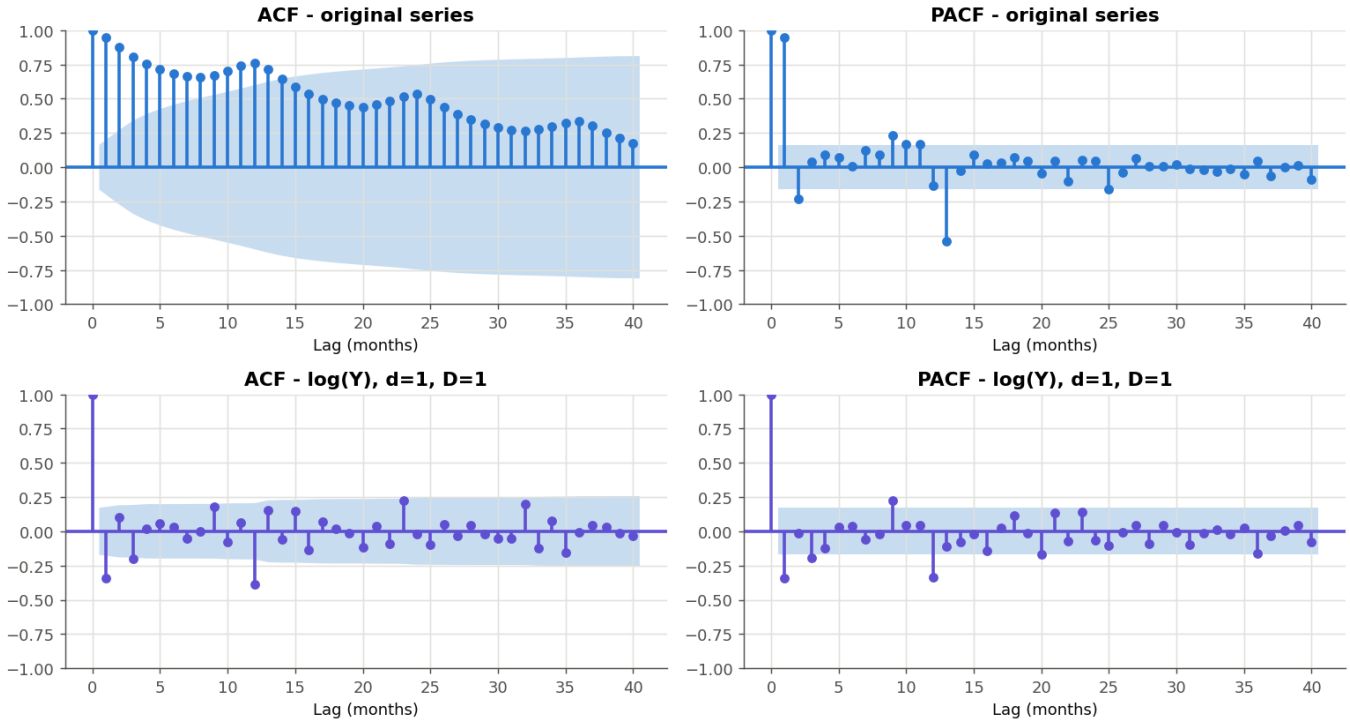


Figure 9: ACF and PACF

- The ACF of the original series decays very slowly, with bumps at 12, 24 and 36. This is the typical signature of a trend plus seasonality (non-stationary).
- For the stationary series log(Y) with d = 1 and D = 1, the largest ACF spikes are at **lag 1 (-0.34)** and **lag 12 (-0.39)**. Smaller spikes at lags 3, 9 and 23 (|r| about 0.2) only just cross the significance band (+/- 0.17). The PACF decays at the start of the series and again around lag 12.
- A single negative ACF spike at lag 1 suggests a non-seasonal MA(1), and a negative spike at lag 12 suggests a seasonal MA(1). The tentative model is **SARIMA(0,1,1)(0,1,1)[12]**.

The AIC grid search over 36 candidate models (p, q = 0..2; P, Q = 0..1) selected the same model:

order	seasonal order	AIC
(0,1,1)	**(0,1,1)[12]**	**-389.01**
(1,1,0)	(0,1,1)[12]	-388.64
(1,1,1)	(0,1,1)[12]	-387.28
(0,1,1)	(1,1,1)[12]	-387.22

Fitted coefficients on 1949-1958: MA(1) theta = -0.342 ($p < 0.001$) and seasonal MA(1) Theta = -0.541 ($p < 0.001$). Both are significant.

The non-seasonal **ARIMA** grid (p, q = 0..3, d = 1) selected ARIMA(3,1,3). The Holt-Winters variant with the lowest AIC was **additive trend + multiplicative seasonality**, undamped (AIC 571.6, against 619.4 for additive seasonality). Its parameters were alpha = 0.364, beta = 0.000 and gamma = 0.636.

7.6 Model comparison (test set: Jan 1959 to Dec 1960, 24 months)

Rank	Model	MAE	RMSE	MAPE (%)
1	**Holt-Winters (add. trend, mul. season)**	**28.98**	**32.49**	**6.39**
2	SARIMA(0,1,1)(0,1,1)[12]	39.44	43.18	8.52
3	Seasonal naive	71.25	77.00	15.52
4	ARIMA(3,1,3)	70.70	92.44	14.09
5	Moving average (12)	77.83	103.22	15.47
6	Naive	115.25	137.33	23.58
6	Simple exponential smoothing	115.25	137.33	23.58

(Saved as `outputs/metrics.csv`. MAE and RMSE are in thousand passengers.)

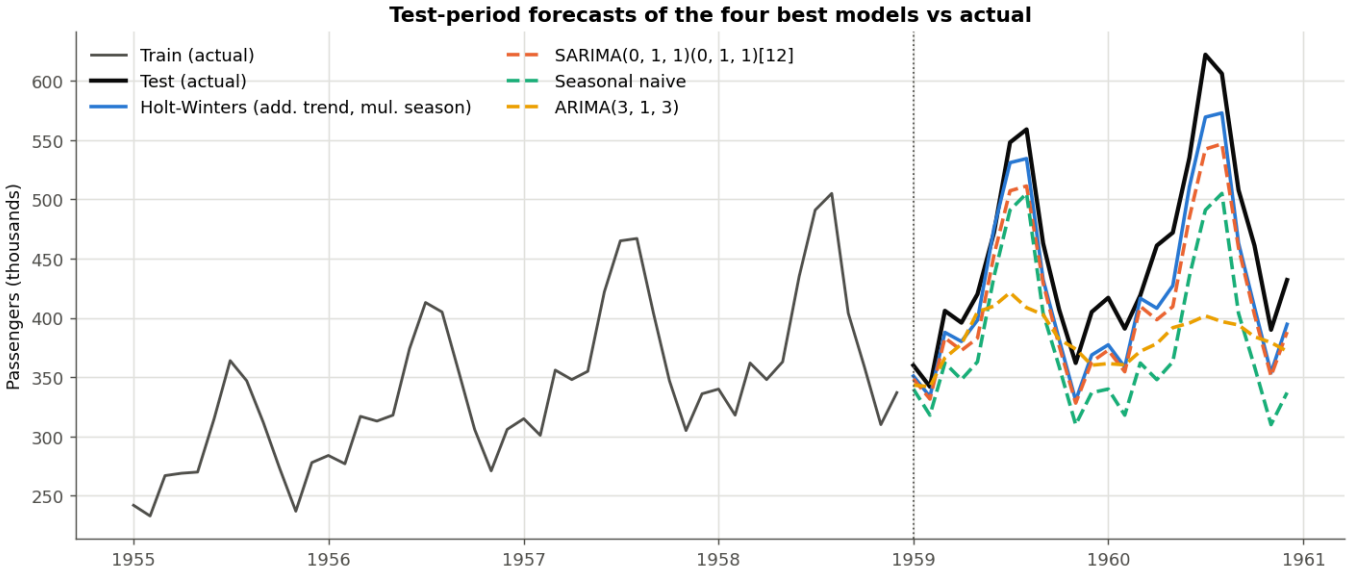


Figure 10: Forecasts vs actual

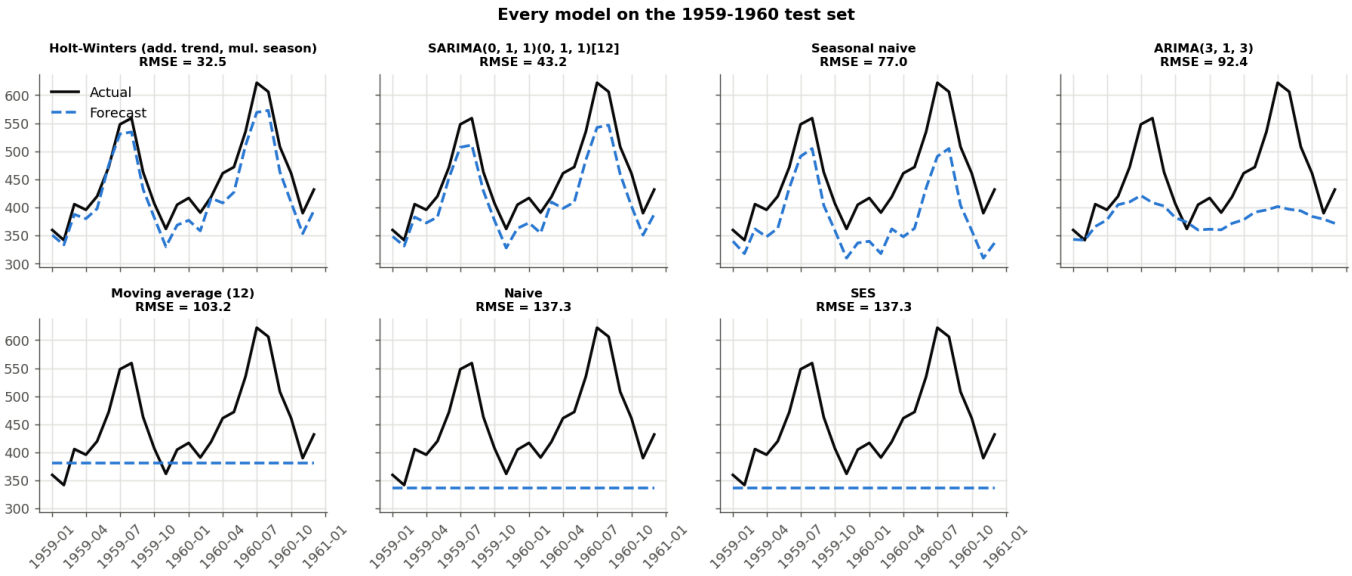


Figure 11: All models on the test set

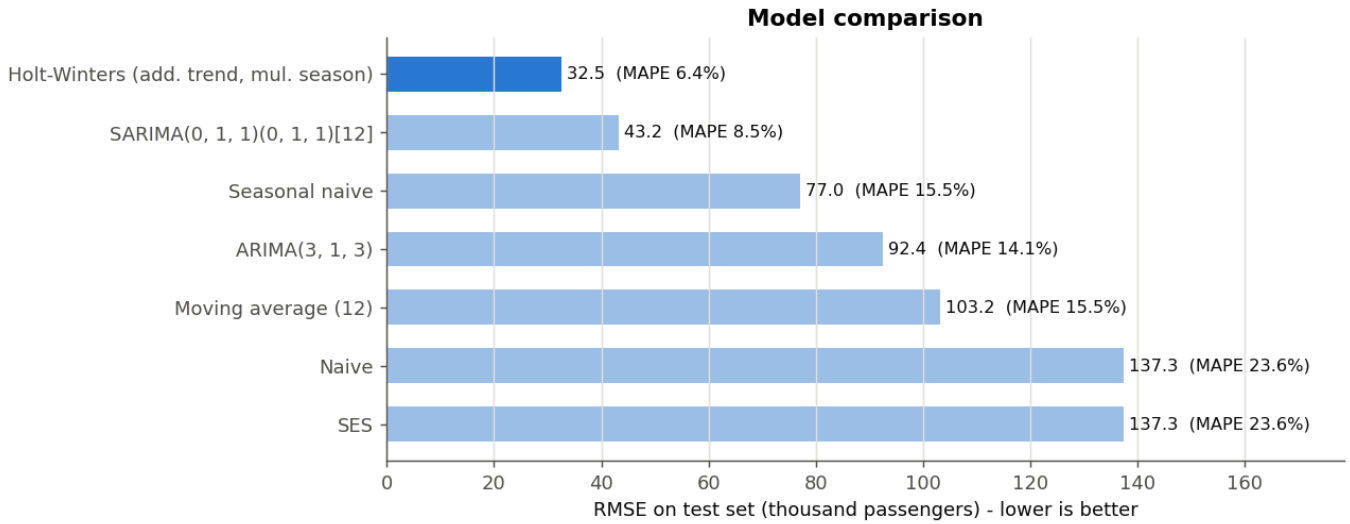


Figure 12: RMSE comparison

Interpretation

- **Holt-Winters** is the most accurate model. Its average error is about 29 thousand passengers per month (6.4 %), less than half the seasonal-naive error.
- **SARIMA** is second (8.5 %). It reproduces the seasonal shape almost perfectly but sits slightly below the actual values. Growth in 1958 was weak (only +3.4 %), so a model trained up to 1958 projects a slower trend than the 11-12 % growth that followed in 1959-60.
- Models **without a seasonal component** (naive, SES, moving average, plain ARIMA) produce flat or smooth lines and cannot follow the summer peaks, so their errors are 2-4 times larger. SES estimated alpha = 1, which makes it identical to the naive forecast.
- Every seasonal model beats every non-seasonal one. Modelling seasonality is the single most important step for this data.

7.7 Residual diagnostics

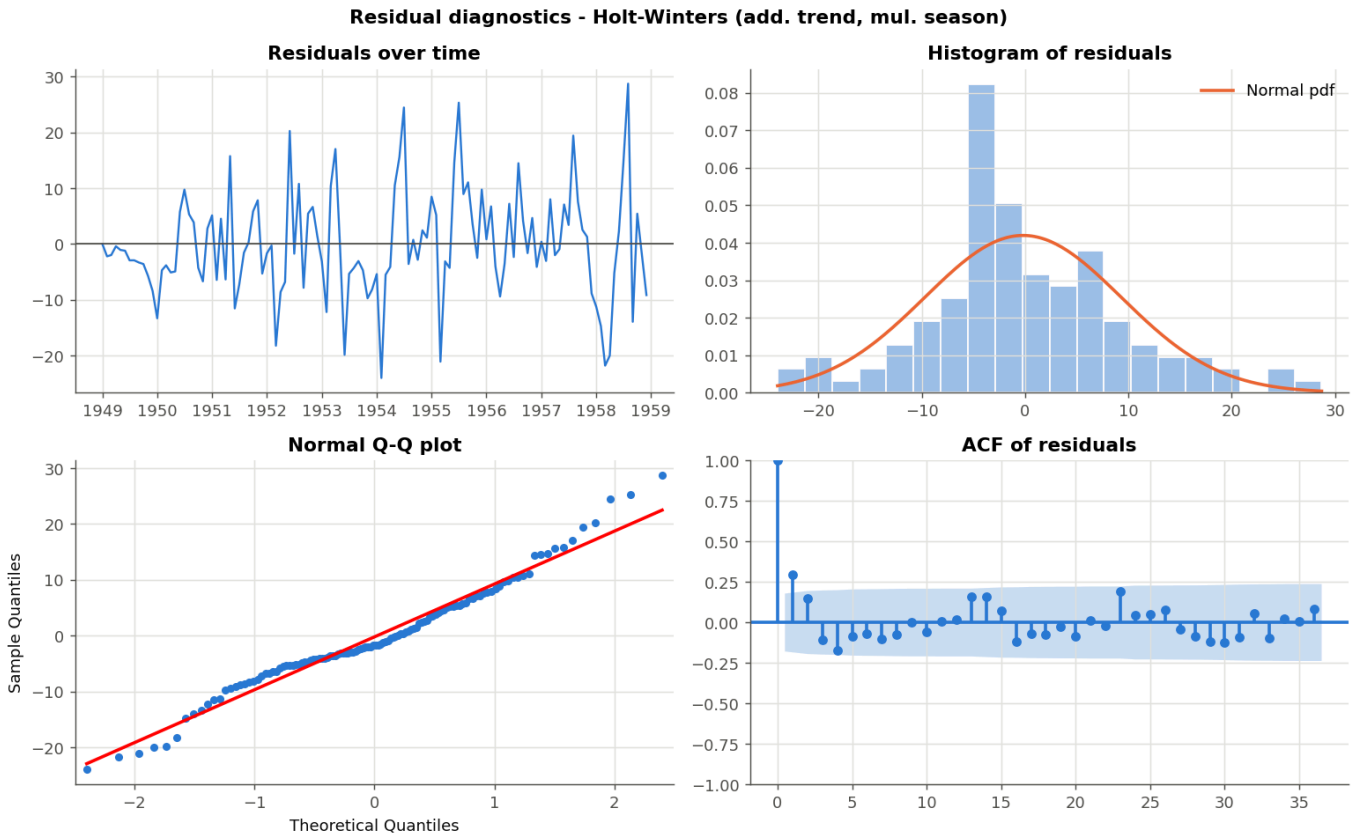


Figure 13: Residual diagnostics - Holt-Winters

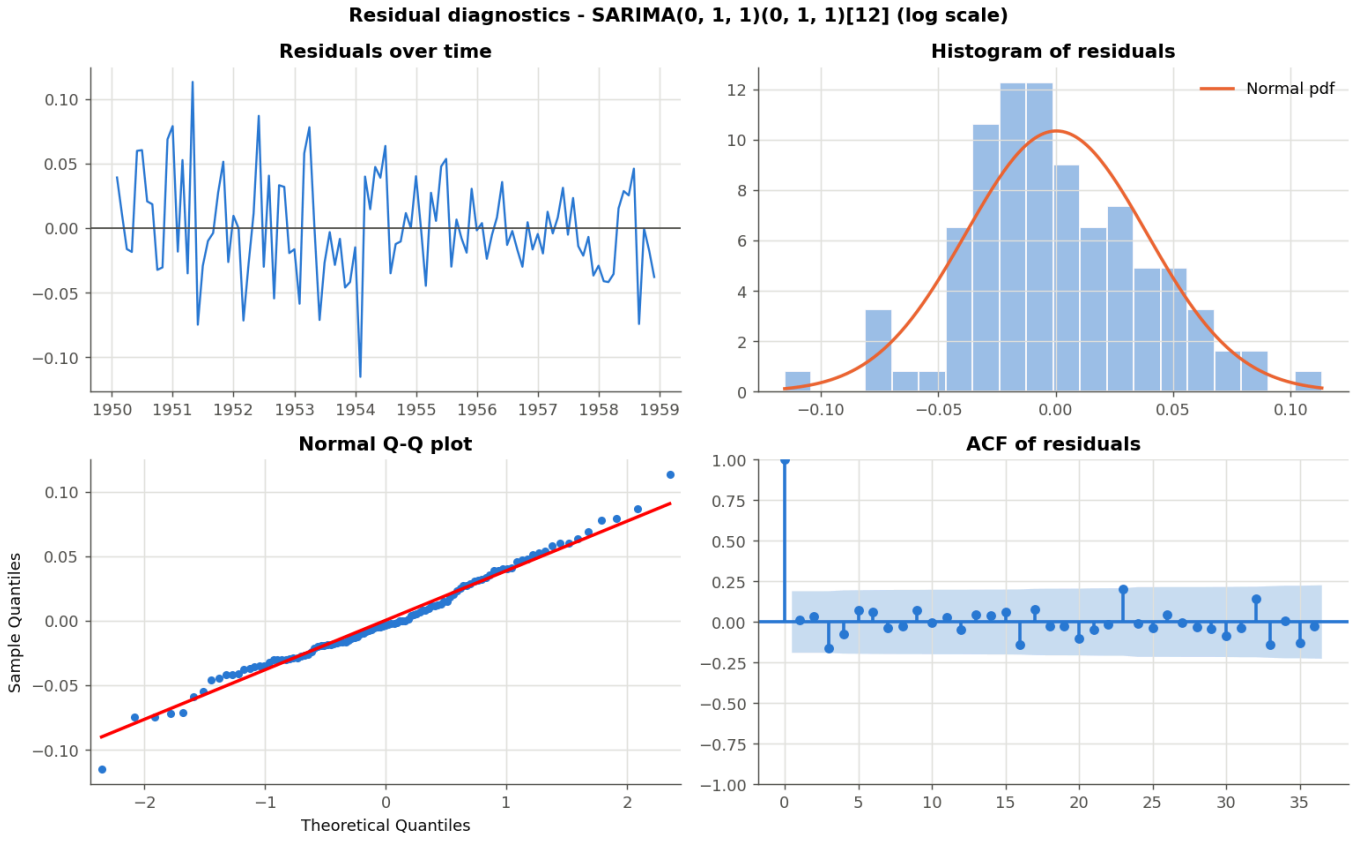


Figure 14: Residual diagnostics - SARIMA

Ljung-Box p-value at lag	6	12	18	24	Jarque-Bera p	Shapiro-Wilk p
Holt-Winters	0.002	0.027	0.012	0.015	0.075	0.023
SARIMA(0,1,1)(0,1,1)[12]	0.559	0.912	0.914	0.804	0.578	0.487

- **SARIMA residuals are white noise.** All Ljung-Box p-values are > 0.05, no ACF bar is significant beyond chance, the Q-Q plot follows the line, and both normality tests pass. Statistically, the model is fully adequate.
- **Holt-Winters residuals** have mean about 0 and are close to normal (Jarque-Bera p = 0.075), but Ljung-Box rejects independence ($p < 0.05$). Some short-lag autocorrelation is left, and the residual spread grows with the level because they are on the original scale. The model forecasts well, but its prediction intervals should be read as approximate.

7.8 Forecast for 1961-1962

The best model on the test set (Holt-Winters) was re-estimated on all 144 months and used to forecast 24 months ahead. Its 80 % and 95 % prediction intervals come from 5,000 simulated future paths.

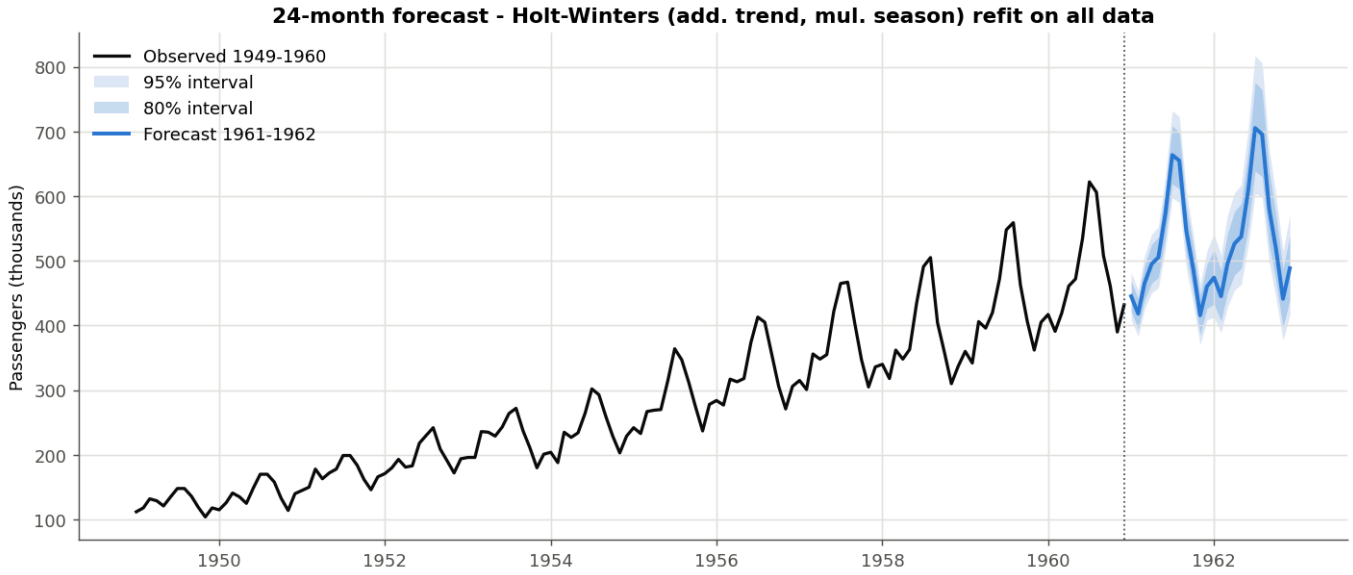


Figure 15: 24-month forecast

Month	Forecast	95 % interval	Month	Forecast	95 % interval
Jan 1961	445	409 - 481	Jan 1962	474	413 - 542
Feb 1961	418	383 - 454	Feb 1962	445	389 - 508
Mar 1961	465	425 - 508	Mar 1962	495	429 - 569
Apr 1961	495	451 - 541	Apr 1962	527	455 - 605
May 1961	506	458 - 553	May 1962	538	464 - 617
Jun 1961	573	518 - 630	Jun 1962	610	525 - 704
Jul 1961	664	597 - 732	Jul 1962	705	605 - 817
Aug 1961	655	590 - 723	Aug 1962	696	598 - 805
Sep 1961	547	489 - 606	Sep 1962	581	499 - 672
Oct 1961	488	436 - 542	Oct 1962	519	444 - 601
Nov 1961	416	371 - 464	Nov 1962	441	377 - 514
Dec 1961	460	409 - 515	Dec 1962	489	416 - 569

- Forecast total for **1961: about 6,132 thousand** passengers (+7.3 % over 1960's 5,714 thousand). For **1962: about 6,518 thousand**.
- The peak forecast is **705 thousand in July 1962**. The seasonal pattern (Jul/Aug high, Nov low) is preserved.
- The intervals widen with the horizon because uncertainty accumulates.

8. Conclusion

1. The airline passenger series has a **strong upward trend** (STL strength 0.996) and a **strong, stable 12-month seasonality** (strength 0.959). The seasonality is **multiplicative**: July/August are about 22 % above average and November about 20 % below.
2. The raw series is **non-stationary** (ADF p = 0.99, KPSS p = 0.01). A **log transform plus one regular and one seasonal difference** makes it stationary (ADF p = 0.0002, KPSS p = 0.10).
3. The ACF/PACF of the stationary series and the AIC grid search both identify the Box-Jenkins **airline model SARIMA(0,1,1)(0,1,1)[12]**. Its residuals pass every diagnostic test (white noise, normal).
4. On the 24-month test set, **Holt-Winters with multiplicative seasonality was the most accurate model (MAE 28.98, RMSE 32.49, MAPE 6.39 %)**. SARIMA was second (MAPE 8.52 %). Both beat the seasonal naive baseline (MAPE 15.52 %) by a wide margin, and every model without seasonality performed poorly.
5. The best statistical fit (SARIMA, clean residuals) is not automatically the best forecaster. Out-of-sample testing is needed to choose a forecasting model.
6. The refitted Holt-Winters model forecasts continued growth of about 7 % in 1961, with a peak of about 705 thousand passengers in July 1962.

Limitations and scope: the dataset is short (12 years) and historical. The forecasts assume that past trend and seasonality continue, and they do not include external factors such as fares, economic conditions or major events. Further work could add rolling-origin cross-validation, Box-Cox transforms, or models such as Prophet or LSTM.

9. References

1. Box, G.E.P., Jenkins, G.M., Reinsel, G.C. (1976/2015). *Time Series Analysis: Forecasting and Control*. Wiley. (Series G: airline passengers)
2. Hyndman, R.J., Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed. OTexts. https://otexts.com/fpp3/
3. Seabold, S., Perktold, J. (2010). *statsmodels: Econometric and statistical modeling with Python*. https://www.statsmodels.org/
4. Dataset CSV: https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv